

Practical Assignment - 2

Department: Computer Engineering & Applications
Course: BCA
Subject: Database Management System Lab (BCAC0816)
Year: 1ST **Semester:** 2ND



Create these tables which consist of following attributes

College

Column Name	Data type	Size
cName	varchar2	10
cstate	varchar2	10
enrollment	int	

Student

Column Name	Data type	Size
sID	int	
sName	varchar2	10
CGPA	number	2,1
marks	int	
DoB	date	

Apply

Column Name	Data type	Size
sID	int	
cName	varchar2	10
major	varchar2	20
decision	char	1

About Database: This database is College Application Database which contain 3 tables.

First one is **Student** that contain information of student such as ID of Student, his name, GPA that he/she scored, size of his High School class i.e. number of students in his/her high school class and student's Date of Birth.

Second table in database is **College** this table contains college information such as name of college, state where it is situated, and its enrollment i.e. number seats in that college.

Third table contains data about applications and this table is named **Apply** each row of this table will contain information about one application. Each application has

- **sID:** This will contain ID (similar to Roll No.) of student who is applying. [Each Application only contain sID of Applicant all other information about student such as his name or GPA can be check from **Student** table]
- **cName:** Name of college where applicant is applying [similar to sID all the other information about college such as its state or enrollment can be retrieve from **College** table]
- **major:** it is the major in which applicant is applying e.g. CS, EE, biology etc.
- **decision:** it is Y or N shows Acceptance or Rejection of Application

Insert the following data to these tables:

Student

sID	sName	CGPA	Marks	DoB
123	Amit	3.9	1000	26-JUN-96
234	Balbir	3.6	1500	7-Apr-95
345	chirag	3.5	500	4-Feb-95
456	Dev	3.9	1000	24-Jul-97
567	Eshan	2.9	2000	21-Dec-96
678	Faizal	3.8	200	27-Aug-96
789	Garvit	3.4	800	8-Oct-96
987	Himani	3.7	800	27-Mar-97
876	Ishan	3.9	400	7-Mar-96
765	Jay	2.9	1500	8-Aug-98
654	Amit	3.9	1000	26-May-96
543	chirag	3.4	2000	27-Aug-98

College

cName	state	enrollment
Stanford	CA	15000
Berkeley	CA	36000
MIT	MA	10000
Cornell	NY	21000
Harvard	MA	50040

Apply

sID	cName	major	decision
123	Stanford	CS	Y
123	Stanford	EE	N
123	Berkeley	CS	Y
123	Cornell	EE	Y
234	Berkeley	biology	N
345	MIT	bioengineering	Y
345	Cornell	bioengineering	N
345	Cornell	CS	Y
345	Cornell	EE	N
678	Stanford	history	Y
987	Stanford	CS	Y
987	Berkeley	CS	Y
876	Stanford	CS	N
876	MIT	biology	Y
876	MIT	marine biology	N
765	Stanford	history	Y
765	Cornell	history	N
765	Cornell	psychology	Y
543	MIT	CS	N

State the SQL Queries for each of the following:

1. Display unique majors.
2. Add column address with data type character and size 20 in Student table.
3. Insert Mathura in Address of student with sID 456.
4. Delete column Address from Student table.
5. List the student names those are having three characters in their Names.
6. List the student names those are starting with 'A'.
7. List the student names those are having third character and fifth char. must be 'e'.
8. List the student names ending with 'y'.
9. List the Students in the order of their CGPA.
10. List the details of the students in order of the ascending of CGPA and descending of DoB.
11. Delete all applications filled at Stanford (*Choose table wisely*)
12. Delete the college Stanford from college table.